Teaching a Digital Humanoid to Walk

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The **Scope**

- Design a **3D virtual humanoid** that learns to stand, balance, and walk using **Reinforcement Learning**.

- Continuous, **feedback loop** training model using a **multi-component reward function** in combination with logic statements.



Training at 20x speed

Background: **Unity Game Engine**



- Powerful Physics Engine
- Allows for **thousands** of high-speed **training episodes**
- Integration of ML-Agents toolkit

Background: **Reinforcement Learning**

- A subfield of machine learning where an **agent learns** a **strategy**. *(Policy)*
- Interacts with its environment through **trial and error.**
- Consists of Four Basic Parts:
  - Environment
  - Agent
  - Action
  - Reward

Goal: **Maximize** a **reward** variable that is calculated by how it interacted with its environment; How **good or bad** its last **action** was.



Three **Important Parts**

Neural Nets

The **“brain”** of the agent

Proximal Policy Optimization

The Optimization method

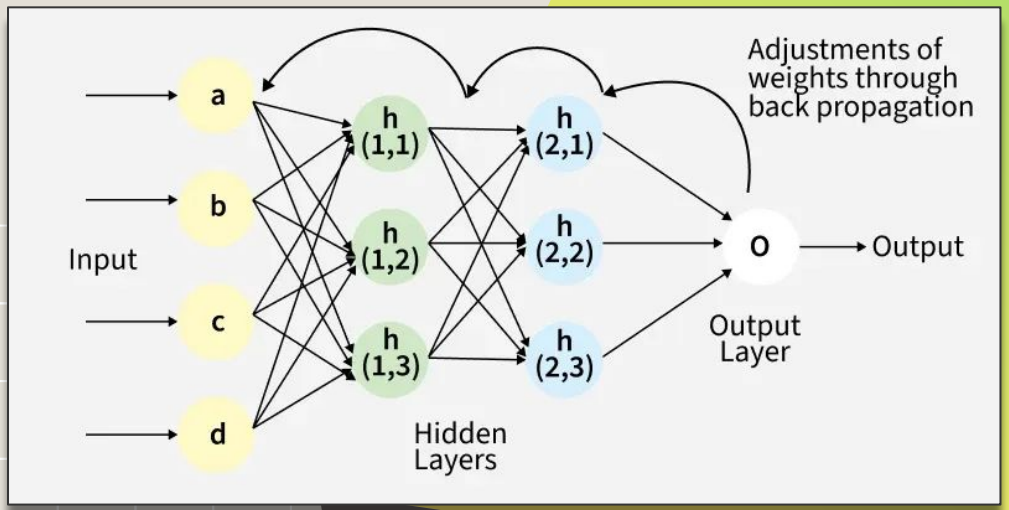
Agent and Critic **feedback loop**

Reward Function

The **guiding function**

What’s “good” and “Bad”

Algorithms: **Neural Nets**



$$Y = wX + B$$

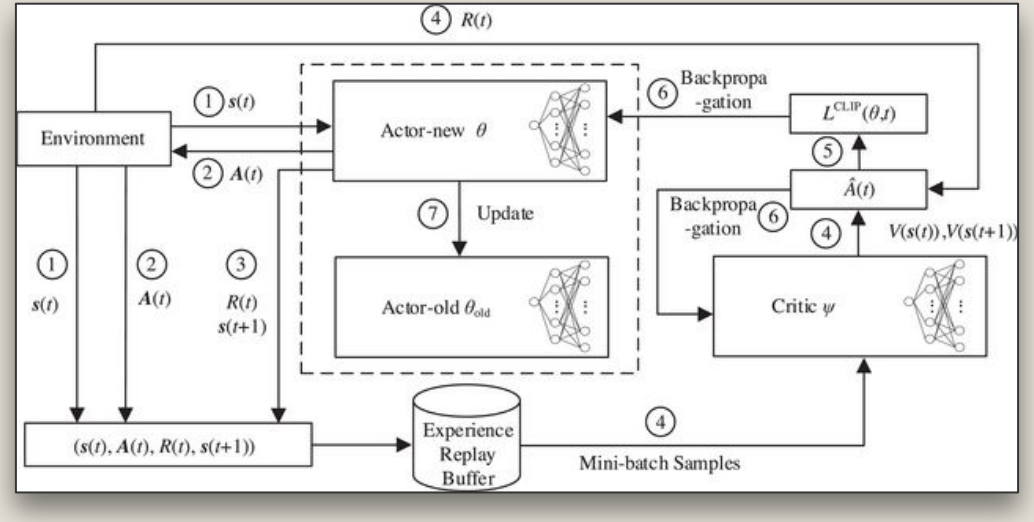
- Processes data through **layers of connected nodes** that learn to **recognize complex patterns**, mathematically adjusting the strength of their connections based on **feedback**.

- The agent's **policy** is a neural network.

Algorithms: **Proximal Policy Optimization**

Core Function: **Prevent** destructively **large policy updates** during training by optimizing a **"clipped"** section of the policy.

Rule: **Limit** the **change** between the old and new policies; Ensures each new iteration stays **close to the previous**, stable policy.



The most important part...

The **Reward Function**

Breaks down the end goal into smaller, simpler objectives.

Staying Upright

A constant, small reward is added when the Agent is upright.

Smooth Movements

Large, erratic movements will cause reward to be lost.

Moving Forwards

A constant reward is awarded when the agent is moving forwards.

Alternating Steps

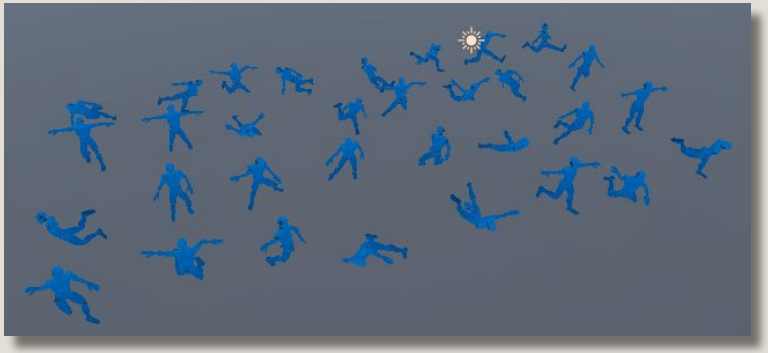
Rewards are added if the agent takes alternating steps.

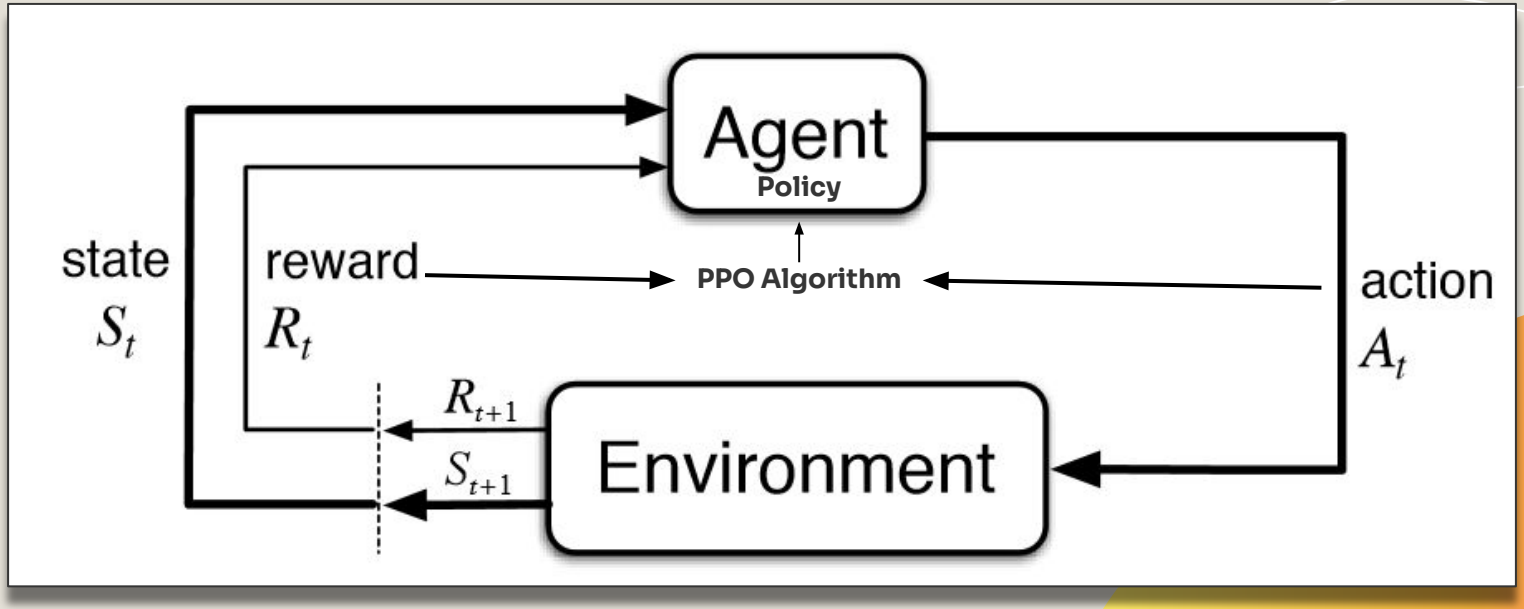
Penalties

If the agent falls, has an improper stance, or doesn't move forwards.

Restart Episode and lose reward.

Methodology





Results

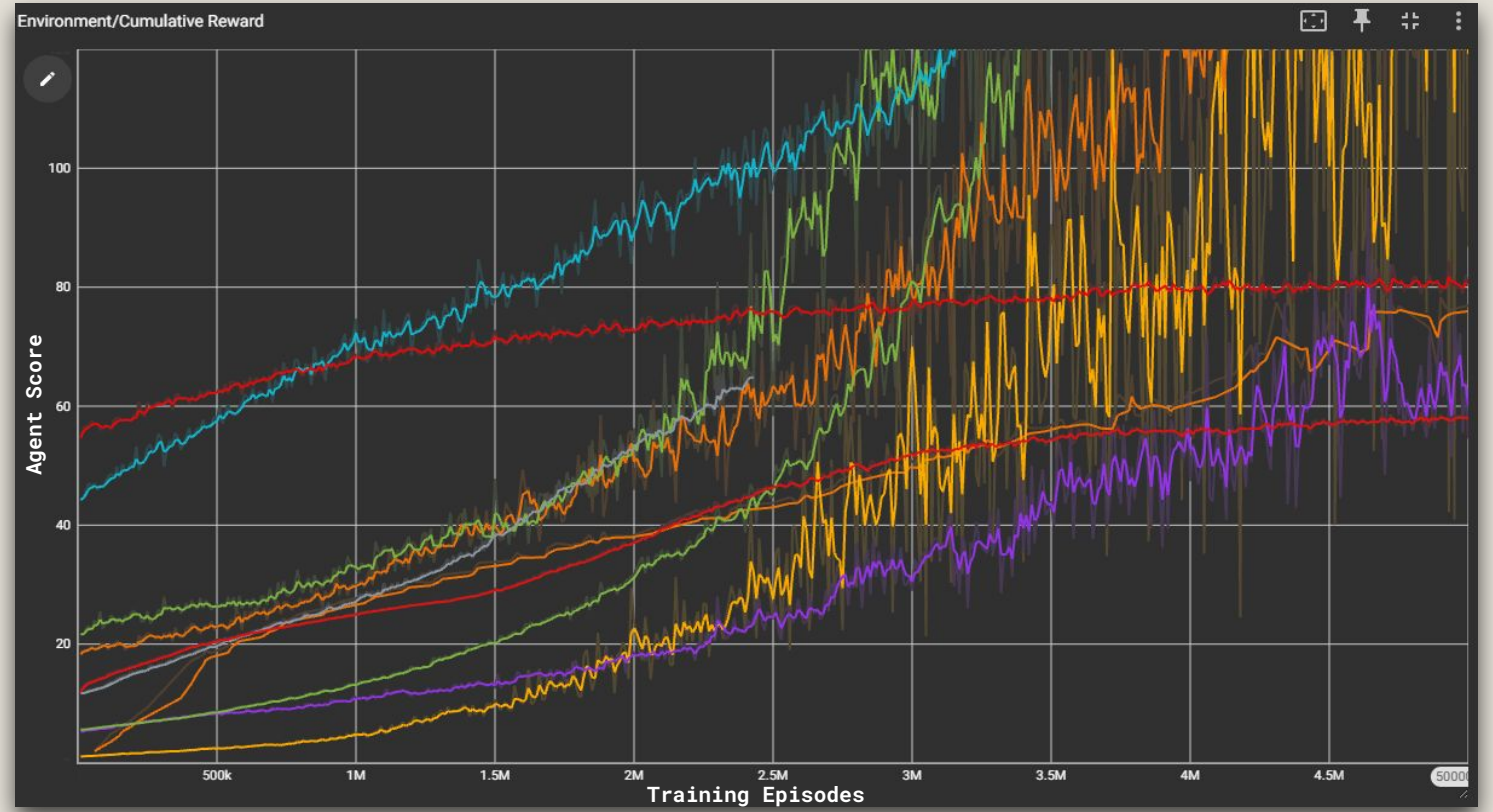
“Standing” after ≈500,000 training episodes



- **Walking**
  - Found **local minima** of shuffling, hopping, or gyrating forwards
  - Agent was reluctant to take first steps

- **Standing**
  - Successfully stood up with intended posture

Results & Demo



Key Points

- Design a **3D virtual humanoid** that learns to stand, balance, and walk.

- Using a complex **Reward Function** to break down the goal into **smaller objectives.**

- Using **Neural Nets** and **Proximal Policy Optimization** to learn what action is optimal for each state the agent is in.



Day one of training...